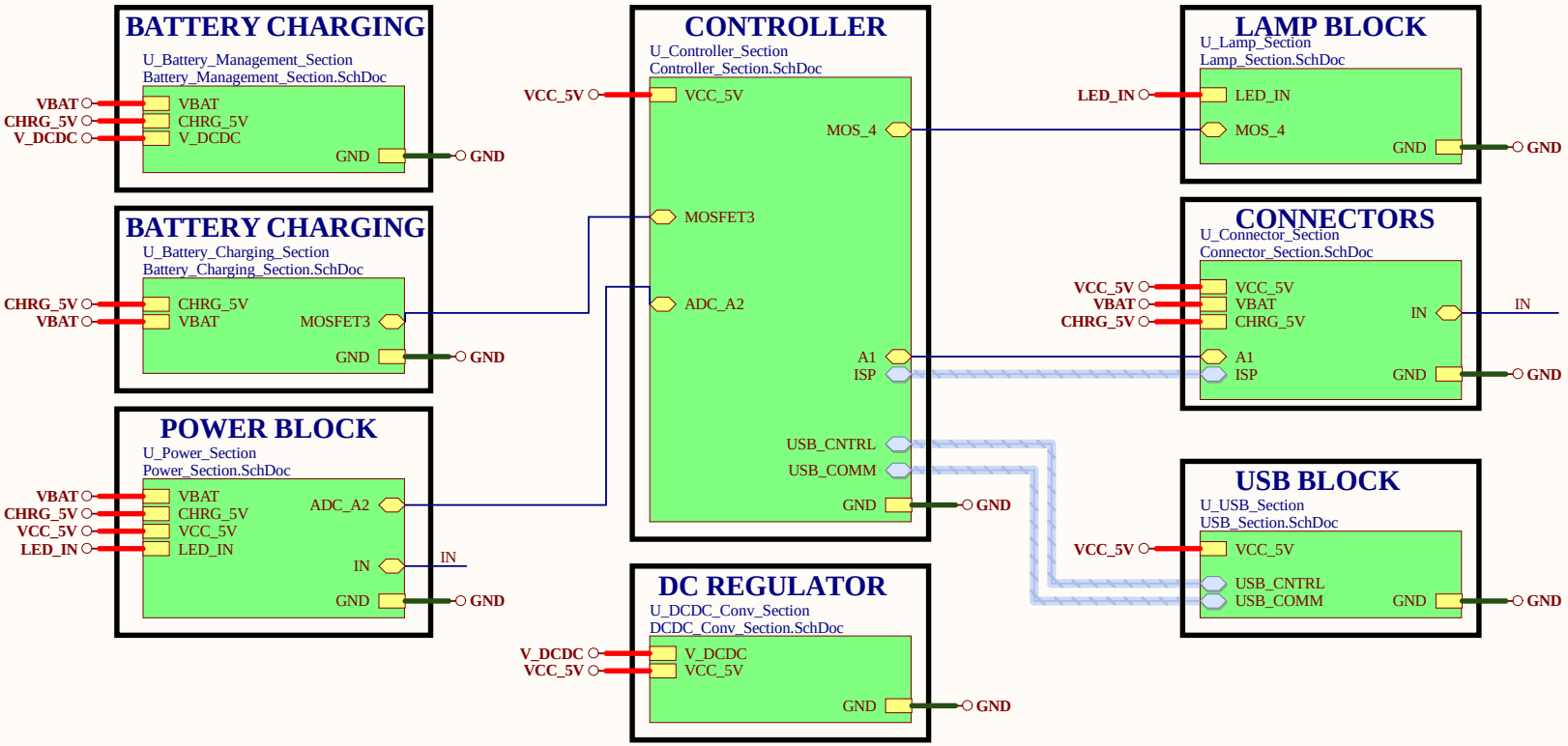
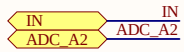
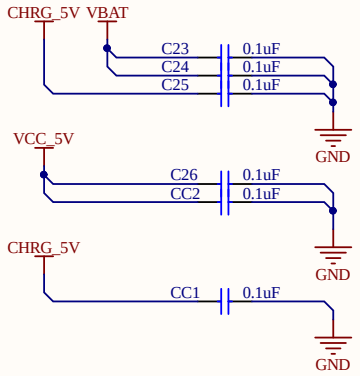
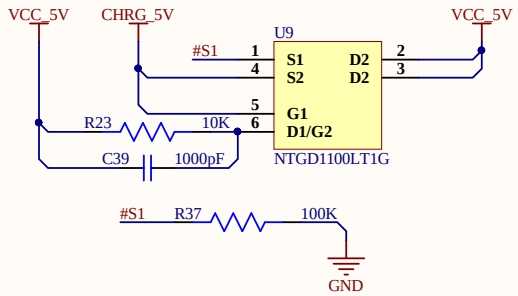

BATTERY CHARGING
U_Battery_Management_Section
Battery_Management_Section.SchDoc
VBAT
CHRG_5V
V_DCDC
GND
BATTERY CHARGING
U_Battery_Charging_Section
Battery_Charging_Section.SchDoc
CHRG_5V
VBAT
MOSFET3
GND
POWER BLOCK
U_Power_Section
Power_Section.SchDoc
VBAT
CHRG_5V
VCC_5V
LED_IN
ADC_A2
IN
GND
CONTROLLER
U_Controller_Section
Controller_Section.SchDoc
VCC_5V
MOS_4
MOSFET3
ADC_A2
A1
ISP
USB_CNTRL
USB_COMM
GND
DC REGULATOR
U_DCDC_Conv_Section
DCDC_Conv_Section.SchDoc
V_DCDC
VCC_5V
GND
LAMP BLOCK
U_Lamp_Section
Lamp_Section.SchDoc
LED_IN
MOS_4
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
VCC_5V
VBAT
CHRG_5V
IN
A1
ISP
GND
USB BLOCK
U_USB_Section
USB_Section.SchDoc
VCC_5V
USB_CNTRL
USB_COMM
GND



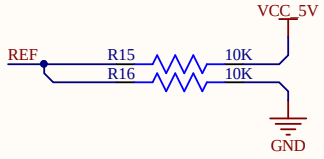
DECOUPLING CAPACITOR



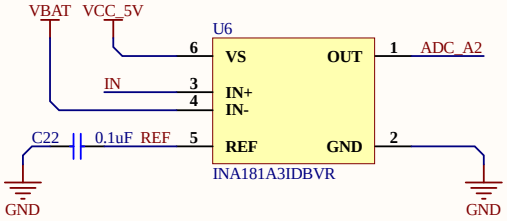
POWER DRIVING IC



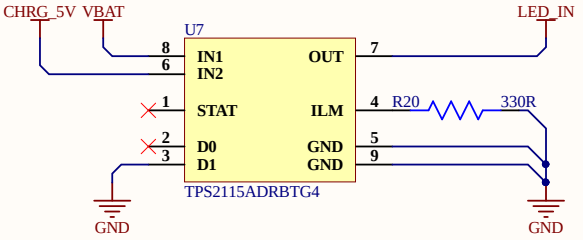
RESISTOR DIVIDER



CURRENT SENSE AMPLIFIER



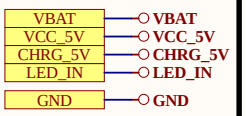
SOURCE SELECTOR



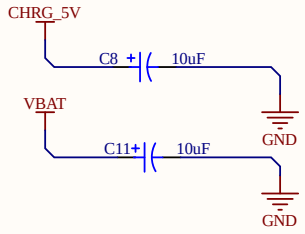
STAT pin can be pulled up to output and connected to MCU for getting IC status

IL=500/RILM

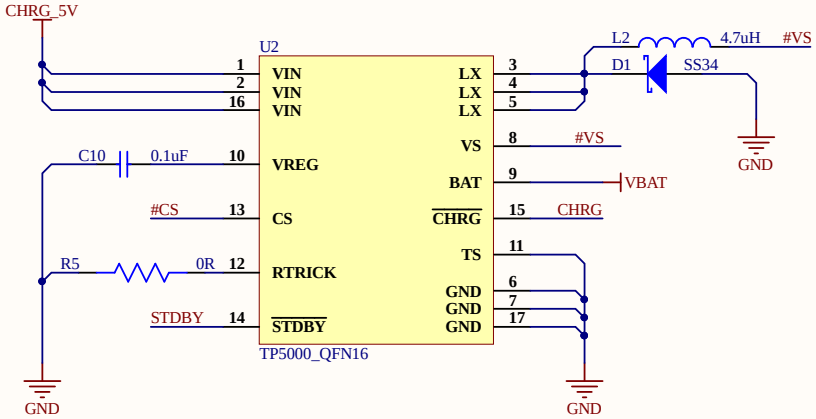
POWER SIGNALS



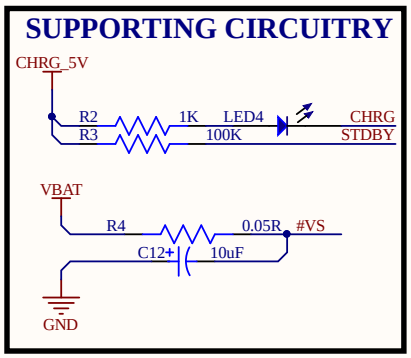
DECOUPLING CAPACITORS



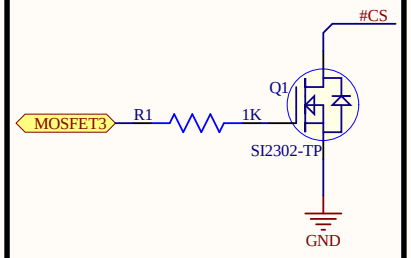
BATTERY CHARGING IC



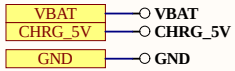
SUPPORTING CIRCUITRY



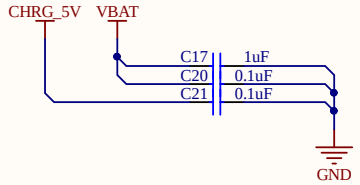
CONTROL CIRCUIT FOR CHARGING/STOP MODE



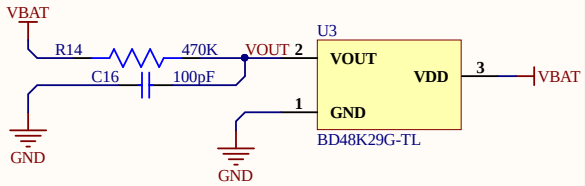
POWER SIGNALS



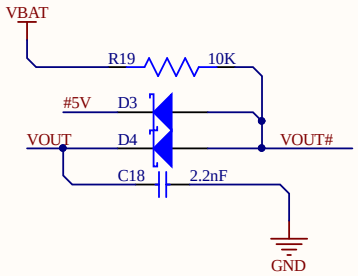
DECOUPLING CAPACITOR



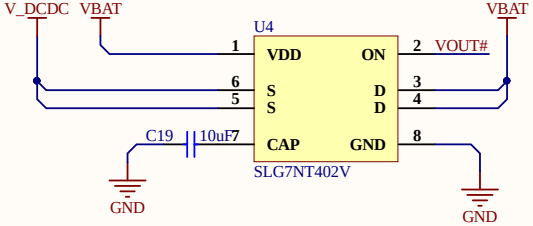
BATTERY SUPERVISORY IC



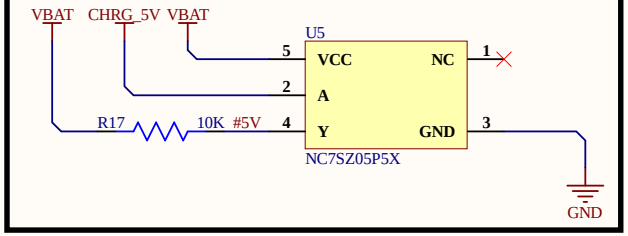
SUPPORTING CIRCUITRY



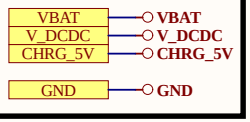
POWER SWITCH



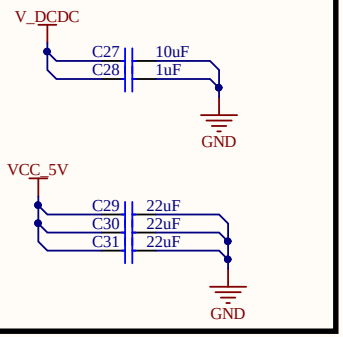
BUFFER INVERTER



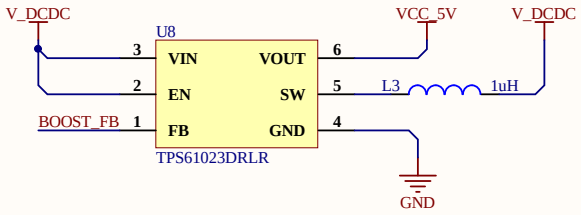
POWER SIGNALS



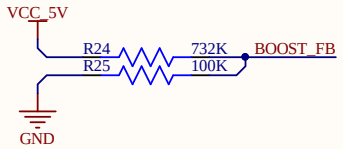
DECOUPLING CAPACITOR



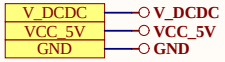
5V BOOST REGULATOR

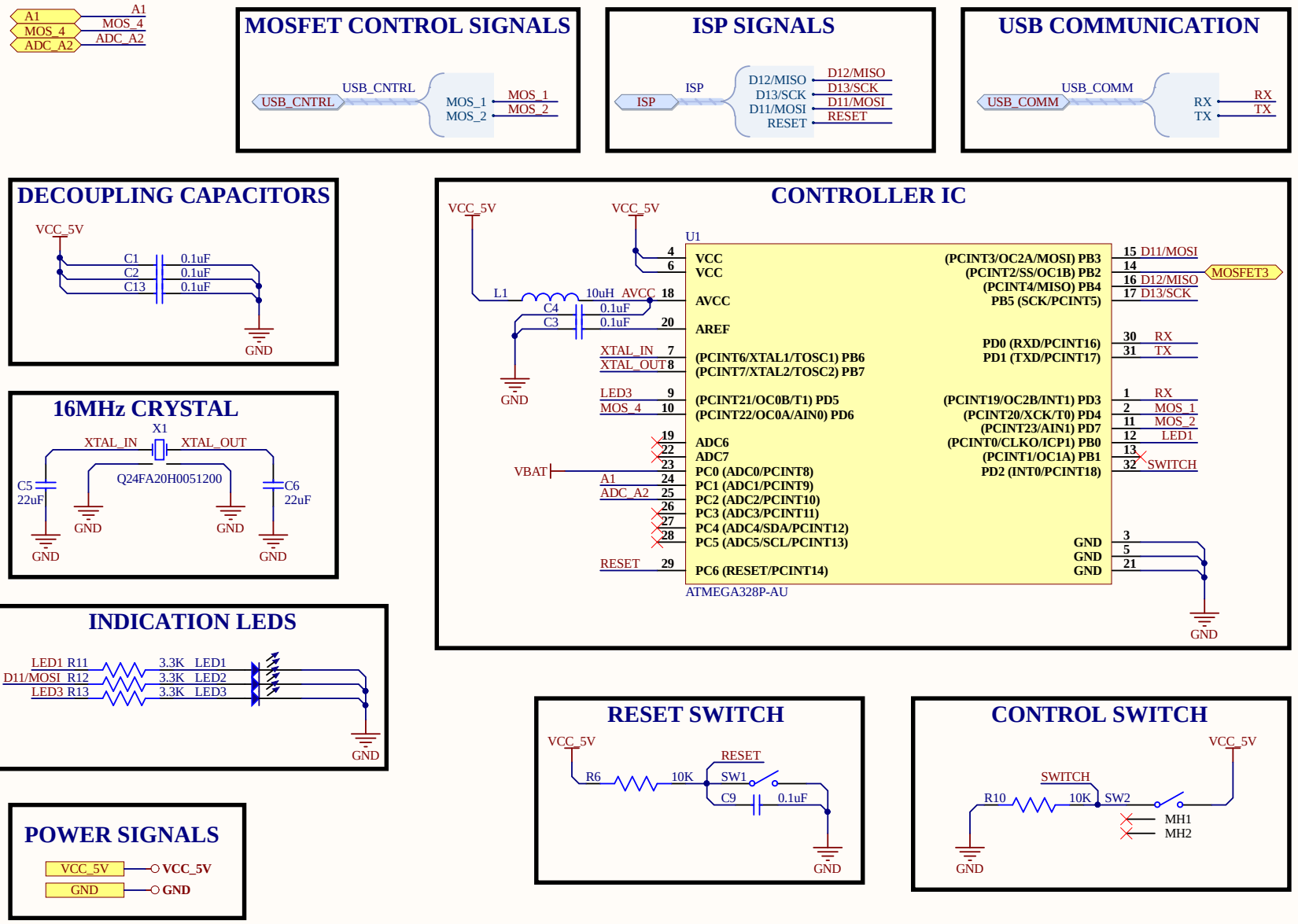


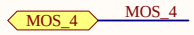
RESISTOR DIVIDER

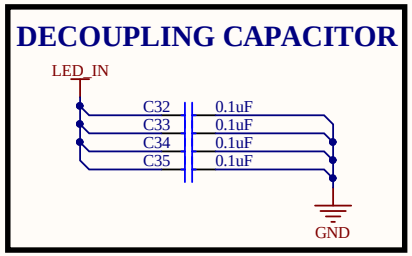


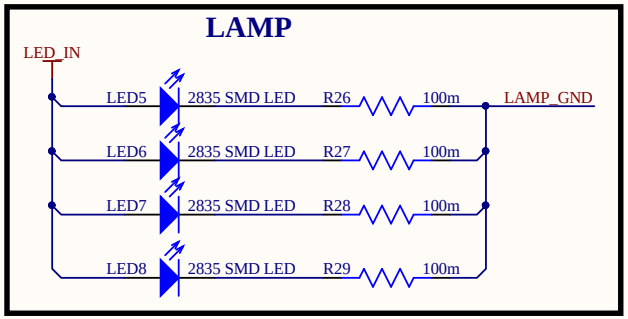
POWER SIGNALS

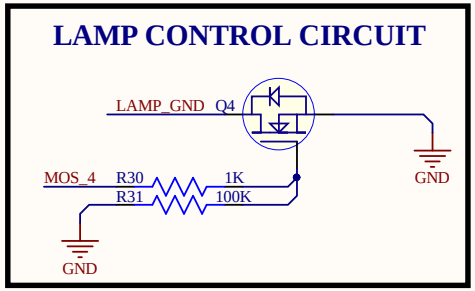


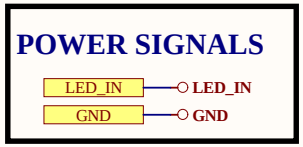

A1
A1
MOS_4
ADC_A2
MOS_4
ADC_A2
MOSFET CONTROL SIGNALS
USB_CNTRL
USB_CNTRL
MOS_1
MOS_2
MOS_1
MOS_2
ISP SIGNALS
ISP
ISP
D12/MISO
D13/SCK
D11/MOSI
RESET
D12/MISO
D13/SCK
D11/MOSI
RESET
USB COMMUNICATION
USB_COMM
USB_COMM
RX
TX
RX
TX
DECOUPLING CAPACITORS
VCC_5V
C1
C2
C13
0.1uF
0.1uF
0.1uF
GND
CONTROLLER IC
VCC_5V
VCC_5V
U1
4 VCC
6 VCC
L1
10uH
AVCC
18 AVCC
C4
0.1uF
C3
0.1uF
20 AREF
GND
XTAL_IN 7 (PCINT6/XTAL1/TOSC1) PB6
XTAL_OUT 8 (PCINT7/XTAL2/TOSC2) PB7
LED3 9 (PCINT21/OC0B/T1) PD5
MOS_4 10 (PCINT22/OC0A/AIN0) PD6
19 ADC6
22 ADC7
VBAT
23
A1 24 PC0 (ADC0/PCINT8)
ADC_A2 25 PC1 (ADC1/PCINT9)
26 PC2 (ADC2/PCINT10)
27 PC3 (ADC3/PCINT11)
28 PC4 (ADC4/SDA/PCINT12)
PC5 (ADC5/SCL/PCINT13)
RESET 29 PC6 (RESET/PCINT14)
ATMEGA328P-AU
(PCINT3/OC2A/MOSI) PB3 15 D11/MOSI
(PCINT2/SS/OC1B) PB2 14
(PCINT4/MISO) PB4 16 D12/MISO
PB5 (SCK/PCINT5) 17 D13/SCK
MOSFET3
PD0 (RXD/PCINT16) 30 RX
PD1 (TXD/PCINT17) 31 TX
(PCINT19/OC2B/INT1) PD3 1 RX
(PCINT20/XCK/T0) PD4 2 MOS_1
(PCINT23/AIN1) PD7 11 MOS_2
(PCINT0/CLKO/ICP1) PB0 12 LED1
(PCINT1/OC1A) PB1 13
PD2 (INT0/PCINT18) 32 SWITCH
GND 3
GND 5
GND 21
GND
16MHz CRYSTAL
X1
XTAL_IN
XTAL_OUT
Q24FA20H0051200
C5
22uF
C6
22uF
GND
GND
GND
GND
INDICATION LEDS
LED1 R11
D11/MOSI R12
LED3 R13
3.3K LED1
3.3K LED2
3.3K LED3
GND
RESET SWITCH
VCC_5V
RESET
R6
10K
SW1
C9
0.1uF
GND
CONTROL SWITCH
VCC_5V
SWITCH
R10
10K
SW2
MH1
MH2
GND
POWER SIGNALS
VCC_5V
VCC_5V
GND
GND


MOS_4
MOS_4

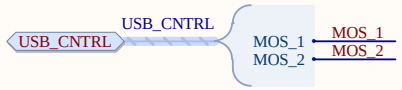

DECOUPLING CAPACITOR
LED_IN
C32
C33
C34
C35
0.1uF
0.1uF
0.1uF
0.1uF
GND


LAMP
LED_IN
LED5
LED6
LED7
LED8
2835 SMD LED
2835 SMD LED
2835 SMD LED
2835 SMD LED
R26
R27
R28
R29
100m
100m
100m
100m
LAMP_GND


LAMP CONTROL CIRCUIT
LAMP_GND
Q4
GND
MOS_4
R30
R31
1K
100K
GND


POWER SIGNALS
LED_IN
GND
LED_IN
GND

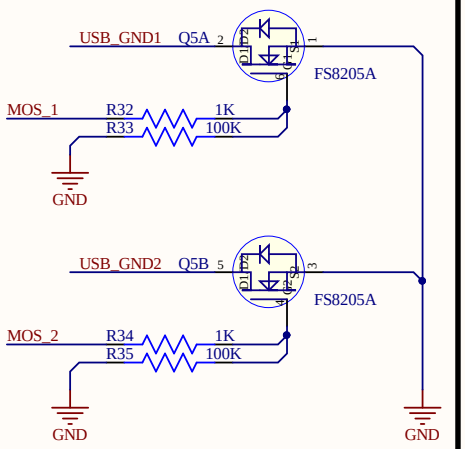
MOSFET CONTROL SIGNALS



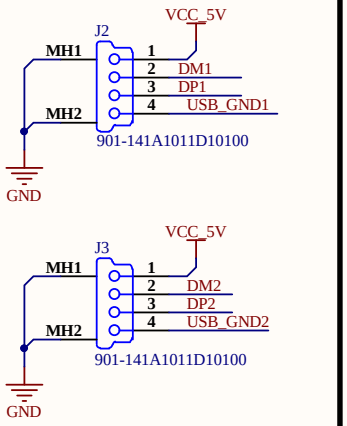
USB COMMUNICATION



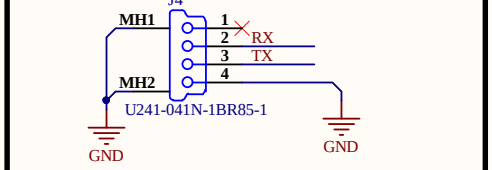
MOSFET CONTROL CIRCUIT



CHARGER OUTPUT



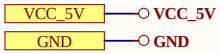
EXTERNAL COMMUNIICATION

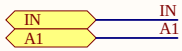


RESISTOR DIVIDER FOR USB DATA LINES



POWER SIGNALS

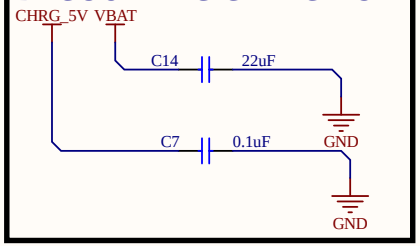




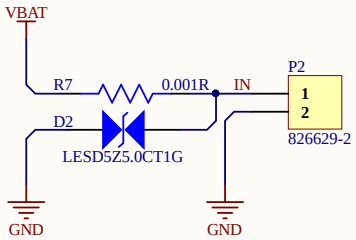
ISP SIGNALS



DECOUPLING CAPACITOR

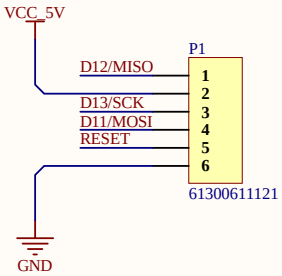


BATTERY HEADER

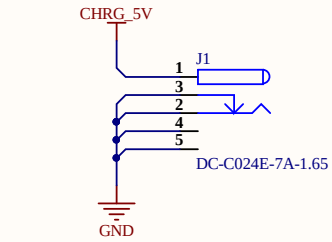


IN signal comes from Current sense amplifier circuitry

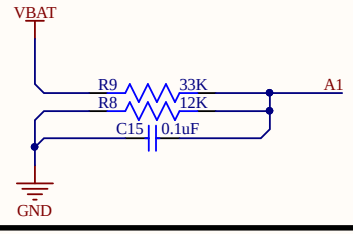
ISP PROGRAMMING HEADER



DC POWER HEADER



FEEDBACK NETWORK



A1 signal comes from Controller

POWER SIGNALS

